Ceph-SRE

An Autonomous LLM Agent for On-Node Ceph Cluster Management

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Agenda

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02 Introduction & Problem

03 Related Work

04 System Architecture

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BACKGROUND

What is Ceph?

The distributed storage system powering modern cloud infrastructure

Distributed Storage

Open-source system that stores data across many machines — block, file, and object workloads in one platform.

Massive Scale

Used in large cloud deployments (OpenStack, CERN, etc). Manages petabytes of data with automatic replication and fault tolerance.

Complex Architecture

Key components: OSDs (storage daemons), MONs (monitors), PGs (placement groups), and RADOS cluster.

Ceph Architecture at a Glance

BACKGROUND

Why Ceph is Hard to Manage

A typical failure diagnosis involves 10+ CLI commands chained manually

Typical operator workflow for a HEALTH_WARN alert:

```
ceph -s
ceph health detail
ceph osd tree
ceph pg dump_stuck
ceph osd df
ceph osd metadata
smartctl
```

⚠ Manual correlation

Operators mentally join dozens of JSON fields to form a hypothesis

⚠ Brittle runbooks

Shell scripts break when Ceph output format changes between releases

⚠ Tribal knowledge

Critical diagnostic steps live in experts' heads, not in code

⚠ Slow response

Simple diagnostics can take 10+ minutes of exploratory CLI work

INTRODUCTION

The Problem

No existing solution combines local LLM inference with deterministic safety to automate Ceph cluster management without exposing data to external APIs.

1

Data Privacy Risk

Existing LLM tools send cluster telemetry to cloud APIs — unacceptable in regulated environments

2

Hallucination Risk

LLMs can suggest destructive commands; no existing system has a formal safety interlock

3

Expertise Bottleneck

Ceph administration requires deep expertise; no tool offers natural-language cluster management

Our Contributions: Ceph-SRE

An autonomous, on-node LLM agent that replaces brittle operator scripts

• Dual-Mode Control Loop

Fast Path (single LLM call) for routine queries + ReAct agent loop for complex multi-step diagnostics

• 4-Tier Safety Framework

Classifies 34 Ceph operations into low/medium/high/critical risk with 100% accuracy — deterministic, not LLM-based

• Threshold Anomaly Detector

Scores cluster health across 7 categories with perfect F1 on synthetic fault scenarios

• Empirical Evaluation

5 LLMs × 245 test cases; 7B Q4 model achieves 85.5% intent accuracy at 6s mean latency

RELATED WORK

Existing Approaches & Limitations

LLM Agents for Cloud Ops

EuroSys 2024 (Chen et al.)

Gap: Targets microservice architectures, relies on cloud-hosted GPT-4. Cannot run on-node, no safety layer, data must leave cluster.

LLM SRE Assistants

ICSE 2023 (Ahmed et al.)

Gap: Focuses on recommendation only — not autonomous execution. Requires human to take every action.

AIOps for Storage

Zhao, He et al.

Gap: Strong on anomaly detection and log analysis; does NOT integrate with autonomous remediation or natural-language interaction.

ReAct Framework

ICLR 2023 (Yao et al.)

Gap: General-purpose Thought-Action-Observation loop. No domain-specific tool registry and no safety interlock for destructive ops.

Ceph Dashboard / cephadm

Ceph Project

Gap: GUI and orchestration — no natural language, no autonomous reasoning, no incident response automation.

Rook (Kubernetes)

CNCF Project

Gap: Automates Ceph lifecycle in K8s but does NOT address interactive diagnostics or real-time operator assistance.

RELATED WORK

Why a New Approach is Needed

Approach	On-node	Natural Language	Autonomous Exec	Safety Layer
GPT-4 Cloud Tools	✗	✓	Partial	✗
Ceph Dashboard	✓	✗	✗	✗
AIOps (log only)	✓	✗	✗	Partial
Rook	✓	✗	Partial	✗
Ceph-SRE (ours)	✓	✓	✓	✓

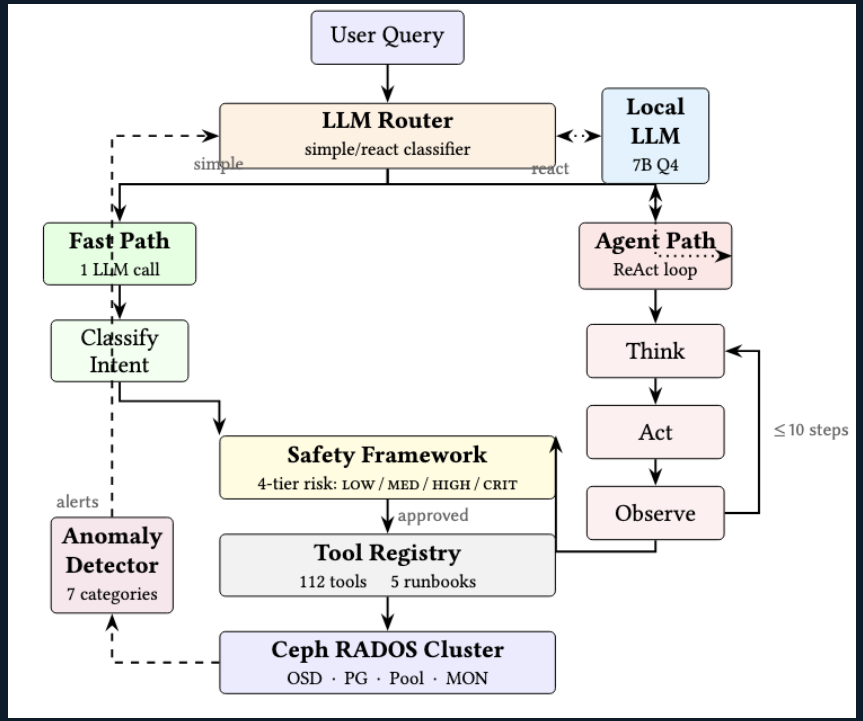
Ceph-SRE is the first system to combine all four properties in a single on-node agent.

PROPOSED METHOD

System Architecture

Four cooperating subsystems — all safety-critical components are deterministic

- **LLM Router** — Classifies each query as *simple* or *react* call; the only routing decision in the system
- **Dual-Path Execution** — Fast Path resolves single-tool queries in; Agent Path runs a ReAct loop up to 10 steps for multi-step diagnostics
- **Safety Framework** — Deterministic 4-tier risk classifier (LOW/MED/HIGH/CRIT)
- **Anomaly Detector** — Threshold-based scoring across 7 categories; runs independently of the LLM in <1ms



PROPOSED METHOD

Dual-Mode Control Loop

Fast Path — Simple Mode

Trigger:
"list pools", "OSD status", "show health"

Trigger:
Single tool call answerable directly

Process:
1. LLM call classifies intent

2. Extract parameters

3. Dispatch to tool

4. Return result

Agent Path — ReAct Mode

Trigger:
"fix degraded PGs", "why is cluster slow"
Multi-step reasoning required

Loop (up to 10 steps, 120s budget):

Think
→ LLM emits Thought + Action + Parameters

Act
→ Safety check → Execute tool

Observe
→ Record output, update scratchpad

Access to 112 tools across 9 categories

PROPOSED METHOD

4-Tier Safety Framework

Risk classification is a dictionary lookup — 100% accurate independent of model quality

LOW	**20 operations** Auto-approve (read-only)
MEDIUM	**8 operations** Log + auto-approve
HIGH	**8 operations** Require explicit operator confirmation
CRITICAL	**4 operations** Confirmation required + full audit log entry

PROPOSED METHOD

Anomaly Detector

Deterministic threshold-based scoring — runs independently of the LLM in < 1ms

Cluster Health Score

$$S = \max(0, 100 - \Sigma\ penalty_i)$$

25 pts — critical anomaly
10 pts — warning
2 pts — info

7 Monitored Categories

- Health
- OSD
- PG
- Capacity
- Balance
- Performance
- Configuration

Score examples: Healthy cluster → 100 · Compound failure (OSD down + near-full + degraded PGs) → 0 · Partial failures → 30-65

PROPOSED METHOD

Case Study: ReAct Loop Diagnosing a Degraded Cluster

"Why is my cluster degraded and how do I fix it?" — qwen2.5:7b, ~30 seconds

Step 1	**Action: get_cluster_health** Obs: status=HEALTH_WARN checks=[PG_DEGRADED, OSD_DOWN]
Step 2	**Action: get_osd_tree** Obs: osd.3 DOWN (host: ceph-2)
Step 3	**Action: get_osd_metadata** Obs: device=/dev/sdc smart_status=FAILED
Step 4	**Action: Final Answer** Obs: osd.3 has failed disk (SMART). Proposed: mark out → await backfill → replace /dev/sdc [HIGH risk → needs confirmation]

Experimental Setup

Hardware

CloudLab node — NVIDIA P100 GPU (12 GB VRAM)
Commodity storage-tier hardware, co-located with Ceph OSD
Ollama serving stack for all model inference

Models Tested

- phi3.5 — 3.8B Q4
- llama3.1:8b — 8B Q4
- qwen2.5:7b — 7B Q4 ★
- qwen2.5:14b — 14B Q4
- llama3.1:8b-fp16 — 8B FP16

Test Suite: 245 cases (multi-run for qwen2.5:7b and llama3.1:8b, n=5 each)

120 queries

Intent Classification

12 intent categories

16 scenarios

ReAct Reasoning

complex diagnostics

20 scenarios

Safety Enforcement

all 4 risk tiers

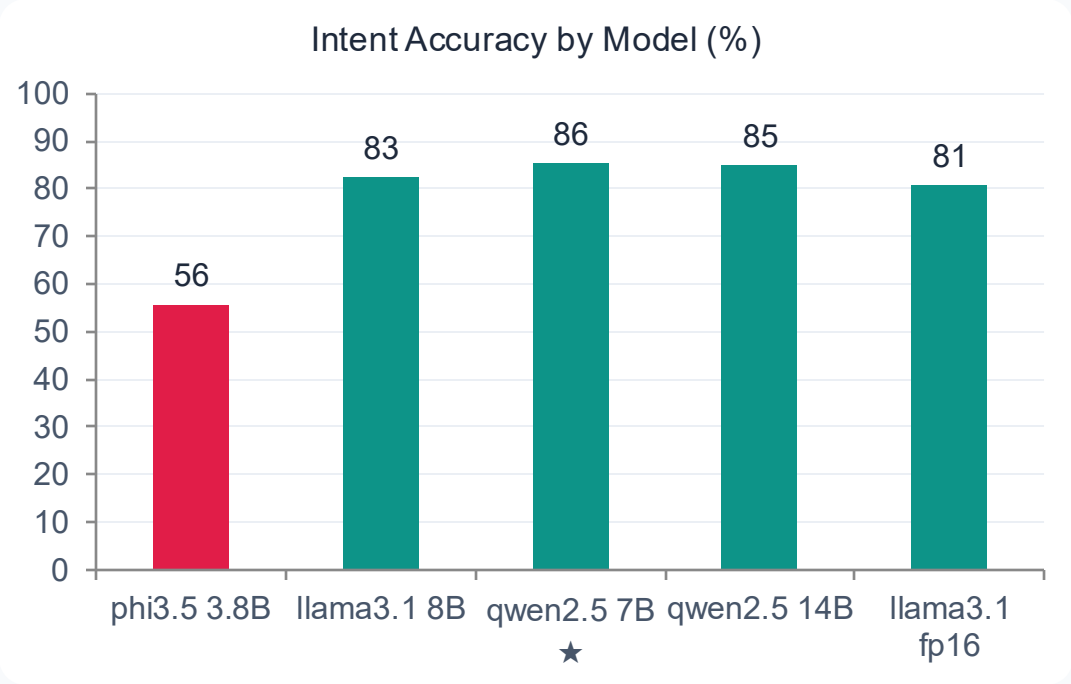
5 states

Anomaly Detection

healthy → compound failure

Results: Intent Classification

120 natural-language queries across 12 categories — which model size crosses the usability threshold?



3.8B fails

phi3.5 scores only 55.8% — below usability threshold. Storage-domain intent needs ≥7B.

7B sweet spot

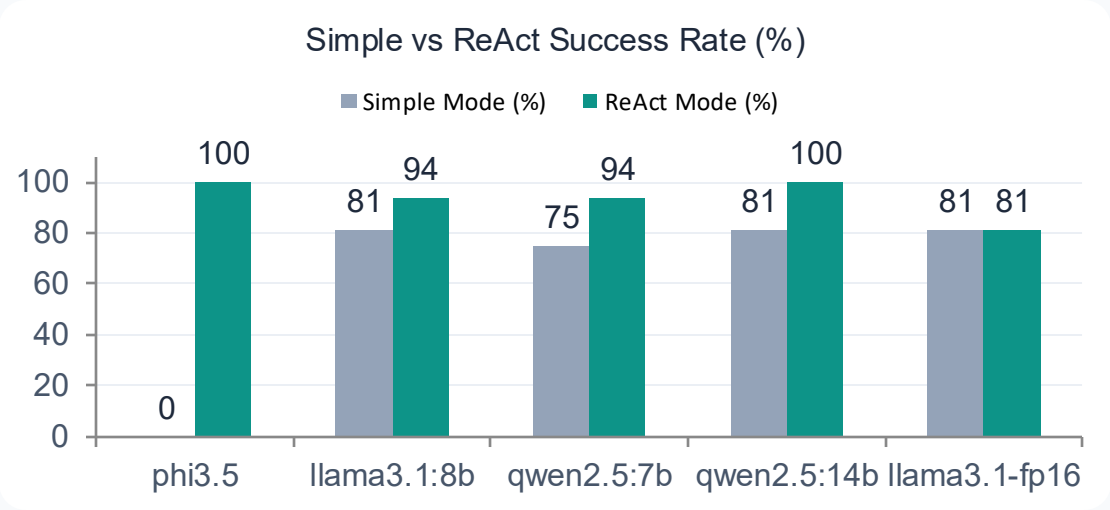
qwen2.5:7b hits 85.5% — scaling to 14B yields NO accuracy gain (+0.5%) but adds 29% latency.

FP16 surprise

Full-precision FP16 (80.8%) underperforms its Q4 counterpart (82.5%), suggesting Q4 preserves instruction-following for structured function calls.

EXPERIMENTS

Results: ReAct Reasoning & Safety



ReAct mode consistently outperforms simple mode for diagnostic tasks (93.8-100% vs 0-81.2%) — validating the dual-path design.

Safety Enforcement

100% Risk Classification

All 5 models — deterministic dictionary lookup, not LLM inference

Safety Confirmation

95% Confirmation Gating

5% gap = policy edge case where medium-risk ops auto-approve

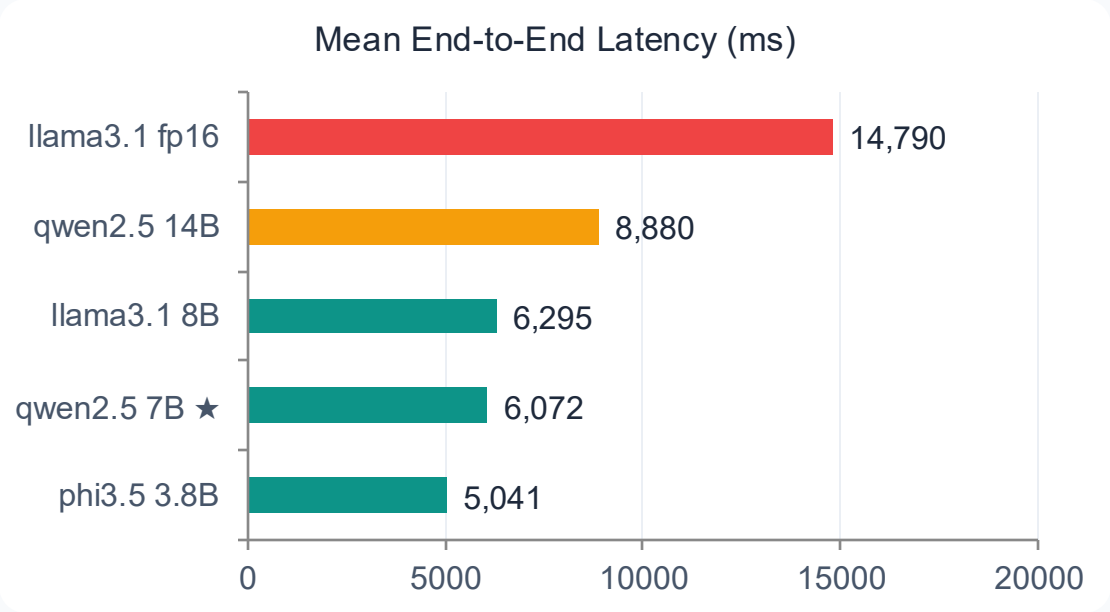
Anomaly Detection

100% F1 Score

All 5 synthetic cluster states correctly scored across all models

Results: Latency Analysis

10 operations × 5 iterations — what is the overhead of adding LLM inference?



7B Q4 = 8-9× CLI
Mean ~6s overhead — acceptable for interactive operator sessions where diagnostics already take minutes manually.

14B exceeds P95 threshold
P95 latency > 10s — crosses the interactive threshold, making 14B impractical for on-node deployment.

FP16 is impractical
~15s mean latency — 21.7× CLI overhead — eliminates FP16 as viable option for production use.

Anomaly detector: <1ms
Runs deterministically, independent of LLM — no latency impact on monitoring pipeline.

CONCLUSION

Key Takeaways

01

7B Q4 is the sweet spot

85.5% intent accuracy at ~6s latency on commodity GPU — sufficient for interactive management

02

Determinism is the key insight

Safety & anomaly components are 100% accurate across ALL model sizes — decouple safety from LLM

03

ReAct vastly outperforms simple mode

93.8-100% vs 0-81.2% for diagnostic tasks — the dual-path design is validated

04

On-node is viable

No internet round-trips, no data leaves the cluster, runs in air-gapped environments

CONCLUSION

Limitations & Future Work

Current Limitations

- Evaluation uses synthetic anomaly scenarios — not live fault injection
- 120-query intent suite doesn't yet cover multi-tenant or RBAC scenarios
- ~15% error rate concentrates in rare management operations (set_osd_out, create_pool)
- LLM router adds one extra round-trip; may misroute highly ambiguous queries

Future Work

- Fine-tune on Ceph-specific corpora to close remaining 15% accuracy gap
- Chaos engineering experiments with live fault injection (OSD failure, network partition)
- Extend to multi-tenant and RBAC-aware scenarios
- Explore smaller models (< 3B) with domain fine-tuning for edge hardware

Ceph-SRE

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Thank You — Questions?

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